formulated only in words, we cannot be sure whether it will work or not. We get a better idea of how feasible such a theory is when it is rephrased in terms of a mathematical model. So far mathematical geneticists who have tried to make the handicap principle into a workable model have failed. This may be because it is not a workable principle, or it may be because they are not clever enough. One of them is Maynard Smith, and my hunch favours the former possibility.

If a male can demonstrate his superiority over other males in a way that does not involve deliberately handicapping himself, nobody would doubt that he could increase his genetic success in that way. Thus elephant seals win and hold on to their harems, not by being aesthetically attractive to females, but by the simple expedient of beating up any male who tries to move in on the harem. Harem-holders tend to win these fights against would-be usurpers, if only for the obvious reason that that is why they are harem-holders. Usurpers do not often win fights, because if they were capable of winning they would have done so before! Any female who mates only with a harem-holder is therefore allying her genes with a male who is strong enough to beat off successive challenges from the large surplus of desperate bachelor males. With luck her sons will inherit their father's ability to hold a harem. In practice a female elephant seal does not have much option, because the harem-owner beats *her* up if she tries to stray. The principle remains, however, that females who choose to mate with males who win fights may benefit their genes by so doing. As we have seen, there are examples of females preferring to mate with males who hold territories and with males who have high status in the dominance hierarchy.

To sum up this chapter so far, the various different kinds of breeding system that we find among animals—monogamy, promiscuity, harems, and so on—can be understood in terms of conflicting interests between males and females. Individuals of either sex 'want' to maximize their total reproductive output during their lives. Because of a fundamental difference between the size and numbers of sperms and eggs, males are in general likely to be biased towards promiscuity and lack of paternal care. Females have two main available counter-ploys, which I have called the he-man and the domestic-bliss strategies. The ecological circumstances of a species will determine whether the females are biased towards one or the other of these counter-ploys, and will also determine how the males respond. In practice all intermediates between he-man and